

A
LAB REPORT
ON
VISUAL PROGRAMMING
Phase VI

By
Kamala Saud
Exam Roll No: 7929006
TU Regd. No:9-2-29-3-2022



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Piyush Pant
Mahendra Ratna Campus

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Chapter One: Basic Programming Concepts

1. C# program to swap two numbers using a third (temporary) variable.

1.1 code

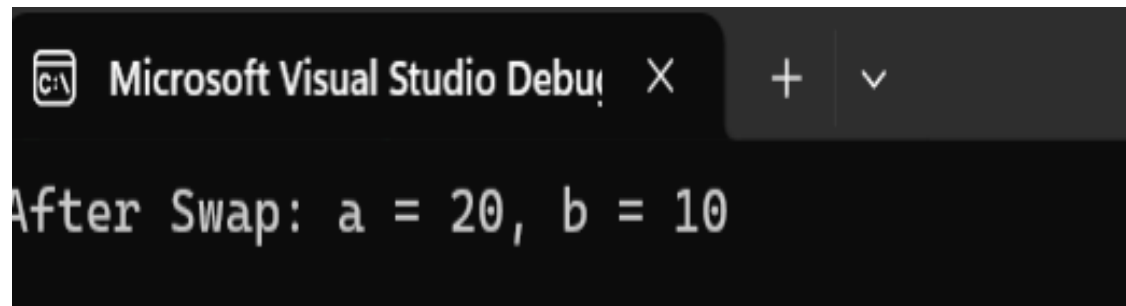
```
using System;

class Program {
    static void Main() {
        int a = 10, b = 20, temp;

        temp = a;
        a = b;
        b = temp;

        Console.WriteLine("After Swap: a = " + a + ", b = " + b);
    }
}
```

1.2 Output



2. Write a C# program to swap two numbers without using any third variable.

2.1 Code

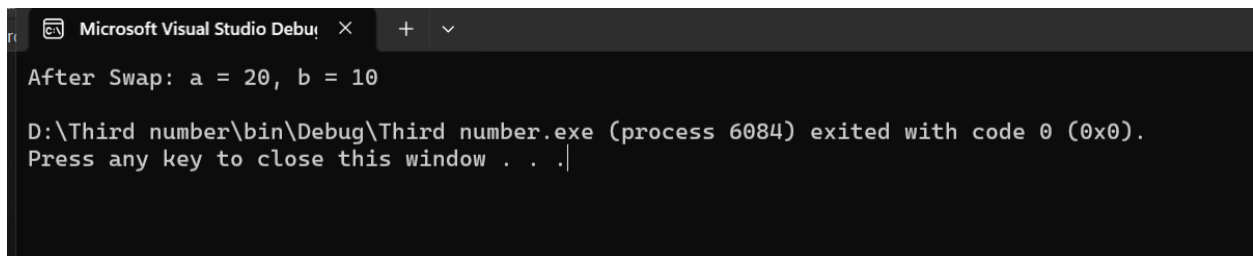
```
using System;

class Program {
    static void Main() {
        int a = 10, b = 20;

        a = a + b;
        b = a - b;
        a = a - b;

        Console.WriteLine("After Swap: a = " + a + ", b = " + b);
    }
}
```

2.2 Output

A screenshot of a Microsoft Visual Studio Debug Console window. The window has a dark background and a title bar that reads "Microsoft Visual Studio Debug Console". The output text is as follows:
After Swap: a = 20, b = 10

D:\Third number\bin\Debug\Third number.exe (process 6084) exited with code 0 (0x0).
Press any key to close this window . . .|

3. Write a C# program that takes a number from the user and checks whether it is a prime number.

3.1 Code

```
using System;

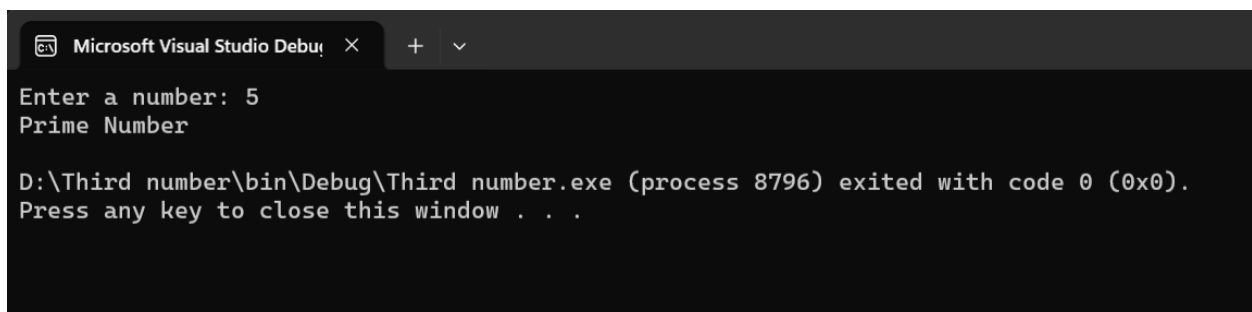
class Program {
    static void Main() {
        Console.Write("Enter a number: ");
        int n = int.Parse(Console.ReadLine());
        bool isPrime = true;

        if (n <= 1) isPrime = false;

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                isPrime = false;
                break;
            }
        }

        Console.WriteLine(isPrime ? "Prime Number" : "Not Prime");
    }
}
```

3.2 Output



```
Microsoft Visual Studio Debug Console
Enter a number: 5
Prime Number

D:\Third number\bin\Debug\Third number.exe (process 8796) exited with code 0 (0x0).
Press any key to close this window . . .
```

4. Write a C# program to check if a given number is a palindrome

4.1 Code

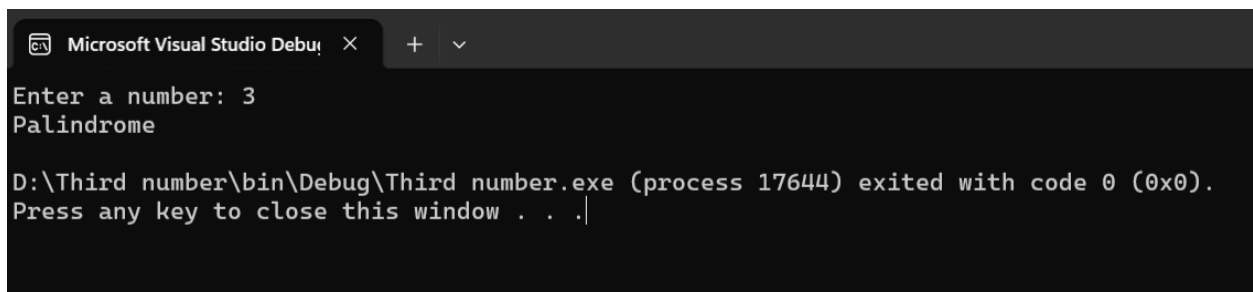
```
using System;

class Program {
    static void Main() {
        Console.Write("Enter a number: ");
        int n = int.Parse(Console.ReadLine());
        int original = n, reverse = 0;

        while (n > 0) {
            reverse = reverse * 10 + n % 10;
            n /= 10;
        }

        Console.WriteLine(original == reverse ? "Palindrome" : "Not
        Palindrome");
    }
}
```

4.2 Output

A screenshot of a Microsoft Visual Studio Debug Console window. The window has a dark background and a title bar that says "Microsoft Visual Studio Debug Console". The output text is as follows:
Enter a number: 3
Palindrome
D:\Third number\bin\Debug\Third number.exe (process 17644) exited with code 0 (0x0).
Press any key to close this window . . .|

5. Write a C# program to reverse a number entered by the user

5.1 code

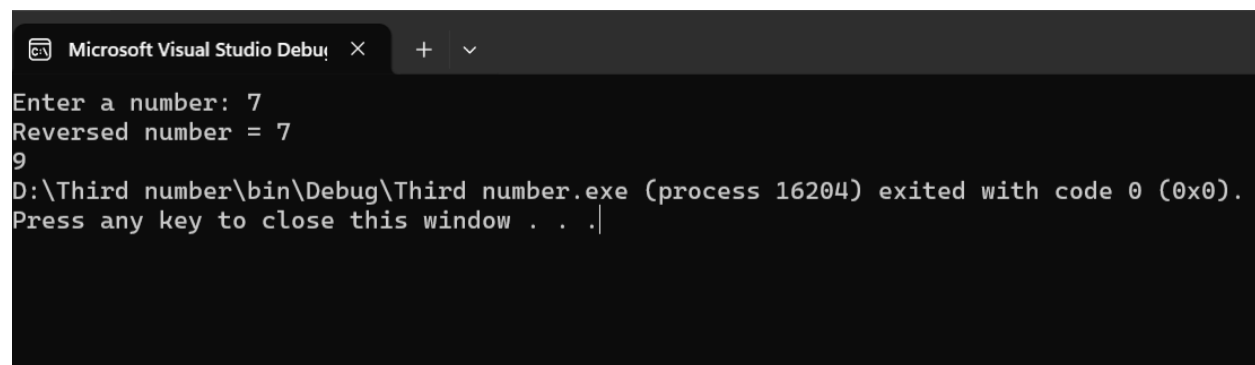
```
using System;

namespace ReverseNumber
{
    class Program
    {
        static void Main()
        {
            Console.Write("Enter a number: ");
            int num = int.Parse(Console.ReadLine());
            int reversed = 0;

            while (num > 0)
            {
                reversed = reversed * 10 + num % 10;
                num /= 10;
            }

            Console.WriteLine($"Reversed number = {reversed}");
            Console.ReadKey();
        }
    }
}
```

5.2 Output

A screenshot of the Microsoft Visual Studio Debug Console. The window title is "Microsoft Visual Studio Debug Console". The output text is as follows:
Enter a number: 7
Reversed number = 7
9
D:\Third number\bin\Debug\Third number.exe (process 16204) exited with code 0 (0x0).
Press any key to close this window . . .|

6. Write a C# program to check whether a number is an Armstrong number

6.1 code

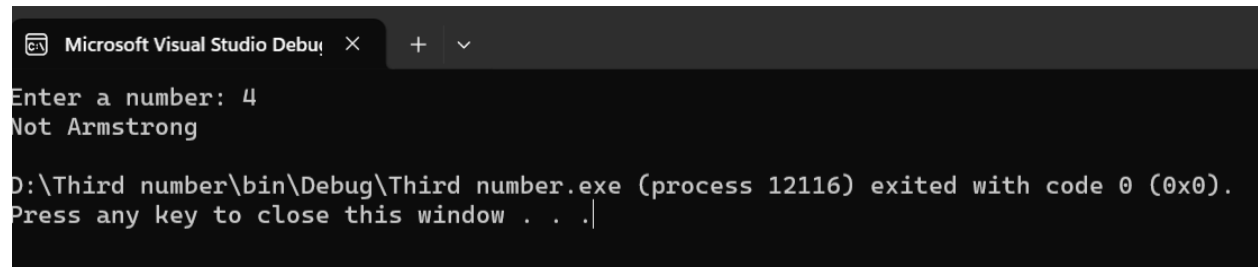
```
using System;

class Program {
    static void Main() {
        Console.Write("Enter a number: ");
        int n = int.Parse(Console.ReadLine());
        int original = n, sum = 0;

        while (n > 0) {
            int digit = n % 10;
            sum += digit * digit * digit;
            n /= 10;
        }

        Console.WriteLine(sum == original ? "Armstrong Number" :
"Not Armstrong");
    }
}
```

6.2 Output

A screenshot of a Microsoft Visual Studio Debug Console window. The window has a dark background and a title bar that says "Microsoft Visual Studio Debug Console". The output text is as follows:
Enter a number: 4
Not Armstrong
D:\Third number\bin\Debug\Third number.exe (process 12116) exited with code 0 (0x0).
Press any key to close this window . . .|

7. Write a C# program to convert temperature from Fahrenheit to Celsius.

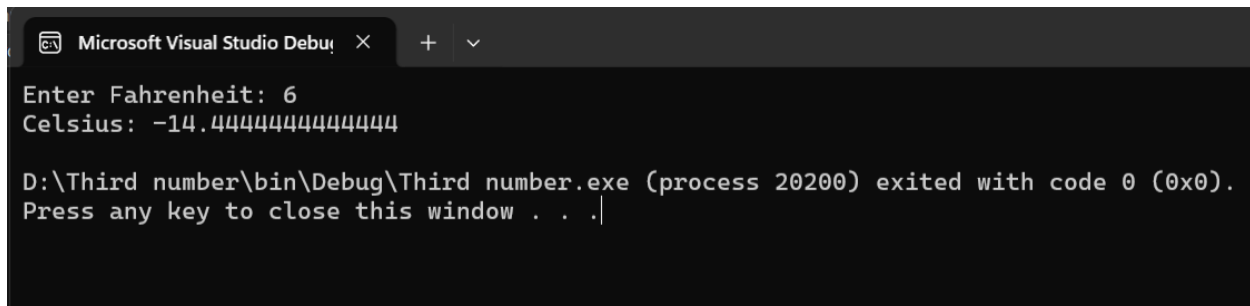
7.1 code

```
using System;

class Program {
    static void Main() {
        Console.Write("Enter Fahrenheit: ");
        double f = double.Parse(Console.ReadLine());

        double c = (f - 32) * 5 / 9;
        Console.WriteLine("Celsius: " + c);
    }
}
```

7.2 Output

A screenshot of a Visual Studio Debug Console window. The window has a dark background and a title bar that says "Microsoft Visual Studio Debug Console". The console output shows the program's execution: it prompts for Fahrenheit input, receives the value 6, calculates the Celsius equivalent as -14.444444444444444, and then displays the result. At the bottom, it shows the process exiting with code 0 (0x0) and prompts the user to press any key to close the window.

```
Enter Fahrenheit: 6
Celsius: -14.444444444444444

D:\Third number\bin\Debug\Third number.exe (process 20200) exited with code 0 (0x0).
Press any key to close this window . . .|
```

8. Write a C# program to print the first N numbers of the Fibonacci series using recursion

8.1 code

```
using System;

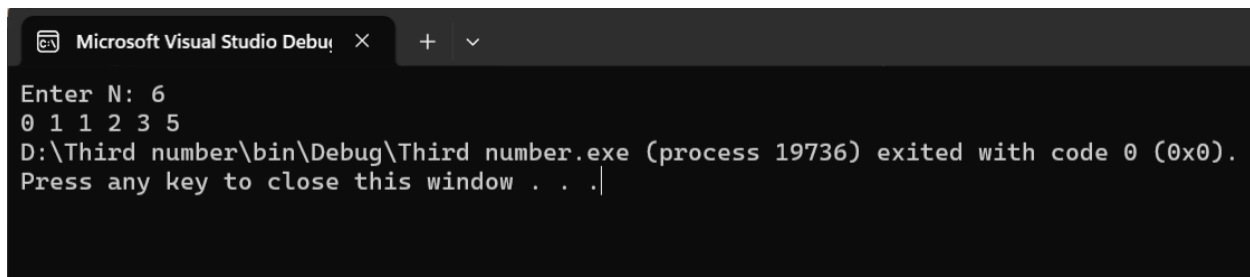
class Program {

    static int Fibonacci(int n) {
        if (n <= 1) return n;
        return Fibonacci(n - 1) + Fibonacci(n - 2);
    }

    static void Main() {
        Console.Write("Enter N: ");
        int n = int.Parse(Console.ReadLine());

        for (int i = 0; i < n; i++)
            Console.Write(Fibonacci(i) + " ");
    }
}
```

8.2 Output

A screenshot of a Visual Studio Debug Console window. The window has a dark background and a title bar that says "Microsoft Visual Studio Debug Console". The output text is as follows:

```
Enter N: 6
0 1 1 2 3 5
D:\Third number\bin\Debug\Third number.exe (process 19736) exited with code 0 (0x0).
Press any key to close this window . . .|
```

9. Write a C# program to find the second largest number in an array.

9.1 code

```
using System;

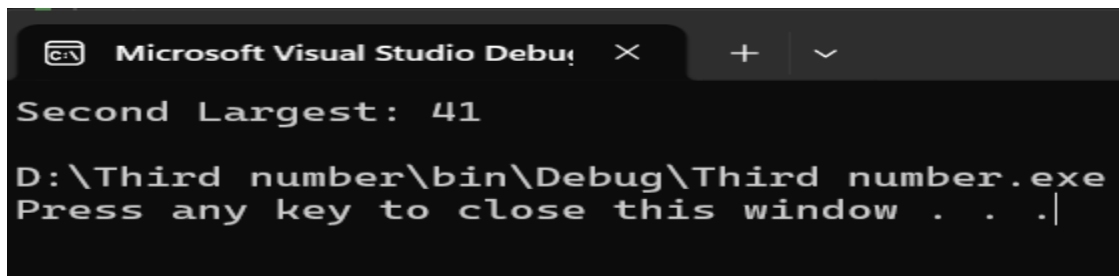
class Program {
    static void Main() {
        int[] arr = { 10, 25, 3, 56, 41 };

        int first = int.MinValue, second = int.MinValue;

        foreach (int n in arr) {
            if (n > first) {
                second = first;
                first = n;
            } else if (n > second && n != first) {
                second = n;
            }
        }

        Console.WriteLine("Second Largest: " + second);
    }
}
```

9.2 Output

A screenshot of a Microsoft Visual Studio Debug console window. The window has a dark background and a title bar that reads "Microsoft Visual Studio Debug". The output text is white and shows "Second Largest: 41" on the first line. The second line shows the file path "D:\Third number\bin\Debug\Third number.exe" and the prompt "Press any key to close this window . . .".

```
Microsoft Visual Studio Debug  X  +  v
Second Largest: 41
D:\Third number\bin\Debug\Third number.exe
Press any key to close this window . . .|
```

10. Write a C# program to count the number of vowels (a, e, i, o, u) in a given string.

10.1 code

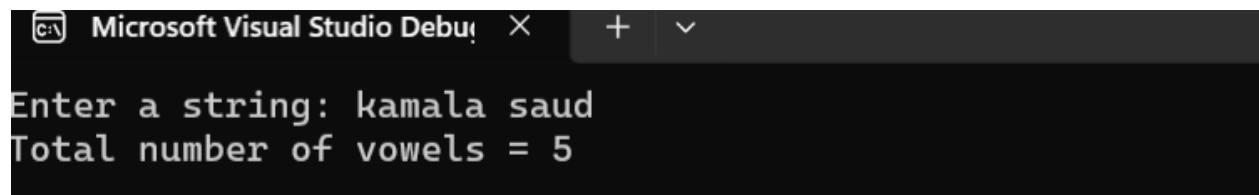
```
using System;

namespace CountVowels
{
    class Program
    {
        static void Main()
        {
            Console.Write("Enter a string: ");
            string str = Console.ReadLine().ToLower();
            int count = 0;

            foreach (char c in str)
            {
                if (c == 'a' || c == 'e' || c == 'i' || c == 'o'
|| c == 'u')
                    count++;
            }

            Console.WriteLine($"Total    number    of    vowels    =
{count}");
            Console.ReadKey();
        }
    }
}
```

10.2 Output

A screenshot of the Microsoft Visual Studio Debug Console window. The window has a title bar with the text "Microsoft Visual Studio Debug" and standard window controls (minimize, maximize, close). The console output shows the program's execution: "Enter a string: kamala saud" followed by "Total number of vowels = 5".

```
Microsoft Visual Studio Debug  X  +  v
Enter a string: kamala saud
Total number of vowels = 5
```

11. Write a C# program to find the missing number in an array that contains numbers from 1 to N (one number is missing).

11.1 code

```
using System;

namespace MissingNumber
{
    class Program
    {
        static void Main()
        {
            Console.Write("Enter value of N: ");
            int n = int.Parse(Console.ReadLine());

            int[] arr = new int[n - 1];
            Console.WriteLine($"Enter {n-1} numbers (one missing from 1 to {n}):");

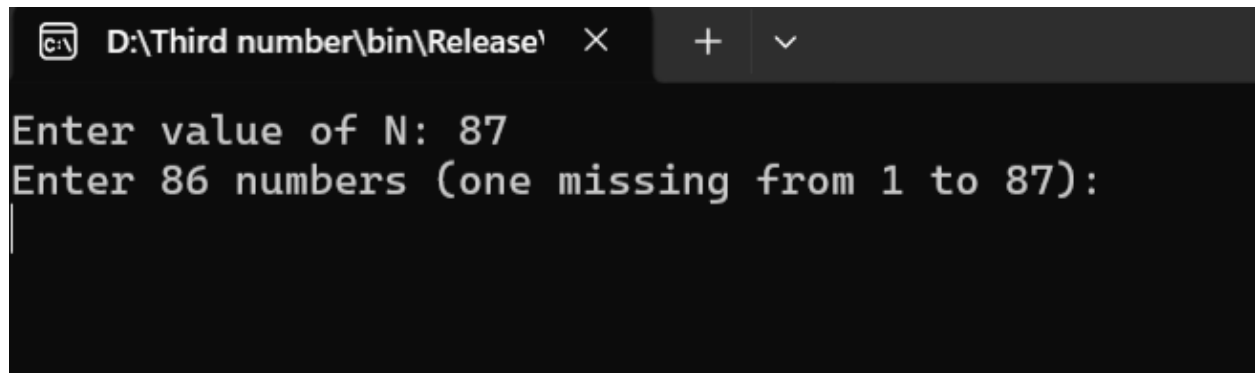
            for (int i = 0; i < n - 1; i++)
                arr[i] = int.Parse(Console.ReadLine());
            int expectedSum = n * (n + 1) / 2;
            int actualSum = 0;
            foreach (int x in arr) actualSum += x;

            int missing = expectedSum - actualSum;

            Console.WriteLine($"Missing number is: {missing}");
            Console.ReadKey();
        }
    }
}
```

```
}  
}
```

11.2 Output



A screenshot of a Windows command prompt window. The title bar shows the file path 'D:\Third number\bin\Release\' followed by standard window controls (minimize, maximize, close). The window content displays the following text in a monospaced font: 'Enter value of N: 87' followed by 'Enter 86 numbers (one missing from 1 to 87):'. A vertical cursor line is positioned at the start of the second line, indicating the program is waiting for input.

```
D:\Third number\bin\Release' X + v  
Enter value of N: 87  
Enter 86 numbers (one missing from 1 to 87):  
|
```

Chapter Two: Strings and Lambda Expressions

12. Write a C# program to reverse a string entered by the user.

12.1 Code

```
using System;

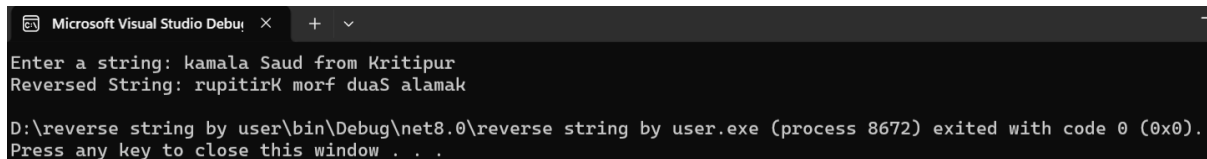
class ReverseStringProgram
{
    static void Main()
    {
        Console.Write("Enter a string: ");
        string input = Console.ReadLine();

        char[] arr = input.ToCharArray();
        Array.Reverse(arr);

        string reversed = new string(arr);

        Console.WriteLine("Reversed String: " + reversed);
    }
}
```

12.2 Output



The screenshot shows a dark-themed window titled "Microsoft Visual Studio Debug Console". The text inside the window is as follows:

```
Enter a string: kamala Saud from Kritipur
Reversed String: rupitirK morf duaS alamak

D:\reverse string by user\bin\Debug\net8.0\reverse string by user.exe (process 8672) exited with code 0 (0x0).
Press any key to close this window . . .
```

13. Write a C# program to remove duplicate characters from a string.

13.1 Code

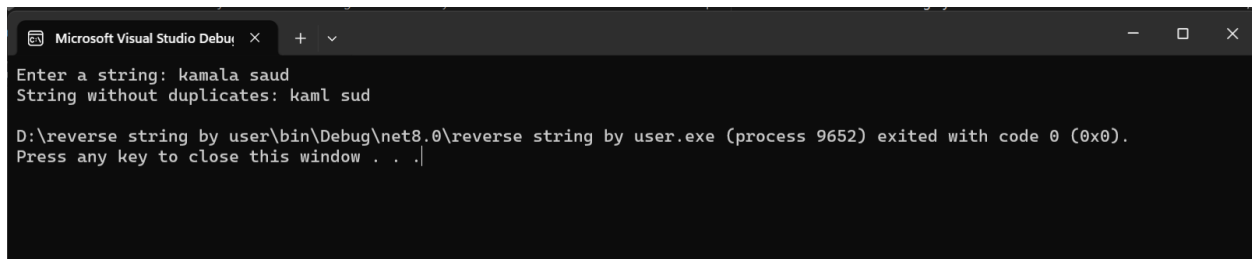
```
using System;
using System.Linq;

class RemoveDuplicates
{
    static void Main()
    {
        Console.Write("Enter a string: ");
        string input = Console.ReadLine();

        string result = new string(input.Distinct().ToArray());

        Console.WriteLine("String without duplicates: " + result);
    }
}
```

13.2 Output

A screenshot of a Microsoft Visual Studio Debug Console window. The window has a title bar with the text "Microsoft Visual Studio Debug" and standard window controls. The console output shows the program's execution: it prompts "Enter a string: kamala saud", displays "String without duplicates: kaml sud", and then shows the process exiting with code 0 (0x0). The prompt "Press any key to close this window . . ." is visible at the bottom.

```
Microsoft Visual Studio Debug x + v
Enter a string: kamala saud
String without duplicates: kaml sud
D:\reverse string by user\bin\Debug\net8.0\reverse string by user.exe (process 9652) exited with code 0 (0x0).
Press any key to close this window . . .
```

14. Write a lambda expression in C# to sort the following list of strings by length:

14.1 code

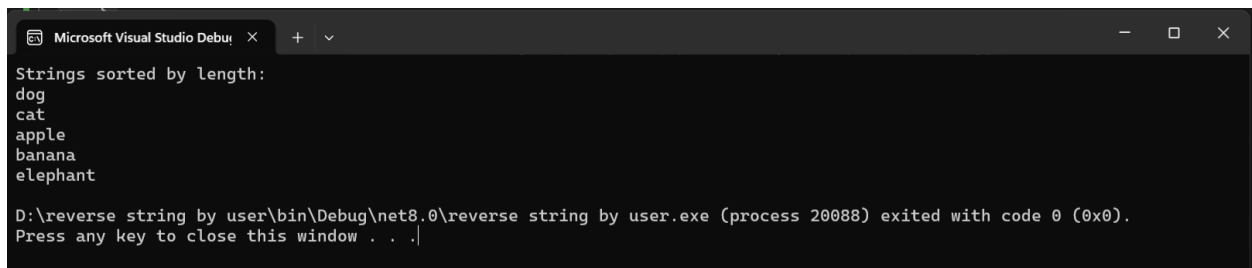
```
using System;
using System.Collections;
using System.Linq;

class SortByLength
{
    static void Main()
    {
        // List of strings
        List<string> words = new List<string> { "apple", "dog",
        "banana", "cat", "elephant" };

        // Sorting using lambda expression
        var sortedList = words.OrderBy(w => w.Length).ToList();

        // Displaying result
        Console.WriteLine("Strings sorted by length:");
        foreach (var word in sortedList)
        {
            Console.WriteLine(word);
        }
    }
}
```

14.2 Output



```
Microsoft Visual Studio Debug  x + v - □ x
Strings sorted by length:
dog
cat
apple
banana
elephant
D:\reverse string by user\bin\Debug\net8.0\reverse string by user.exe (process 20088) exited with code 0 (0x0).
Press any key to close this window . . .|
```

15. C# Program: Check for Negative Numbers Using Lambda Expression

15.1 code

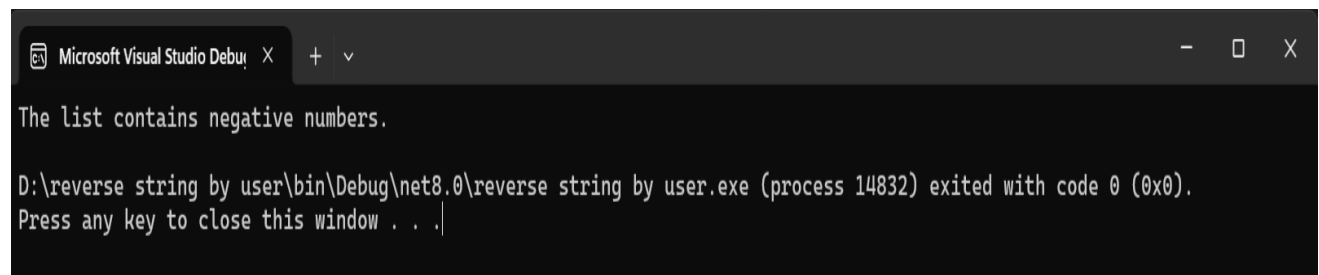
```
using System;
using System.Collections.Generic;
using System.Linq;

class CheckNegative
{
    static void Main()
    {
        // List of integers
        List<int> data = new List<int> { 1, 2, -3, 4 };

        // Lambda expression to check negative numbers
        bool hasNegative = data.Any(n => n < 0);

        // Output result
        if (hasNegative)
        {
            Console.WriteLine("The list contains negative numbers.");
        }
        else
        {
            Console.WriteLine("The list does not contain any negative numbers.");
        }
    }
}
```

15.2 Output

A screenshot of the Microsoft Visual Studio Debug Console window. The window has a title bar with the text "Microsoft Visual Studio Debug" and standard window controls (minimize, maximize, close). The console output shows the text "The list contains negative numbers." followed by a new line. Below that, it shows the command prompt path "D:\reverse string by user\bin\Debug\net8.0\reverse string by user.exe (process 14832) exited with code 0 (0x0)." and the instruction "Press any key to close this window . . .".

```
Microsoft Visual Studio Debug X + v
The list contains negative numbers.

D:\reverse string by user\bin\Debug\net8.0\reverse string by user.exe (process 14832) exited with code 0 (0x0).
Press any key to close this window . . .
```

16. Write a lambda expression in C#

to check if there are any negative numbers in the list:

```
List<int> data = new List<int> { 1, 2, -3, 4 };
```

16.1 Code

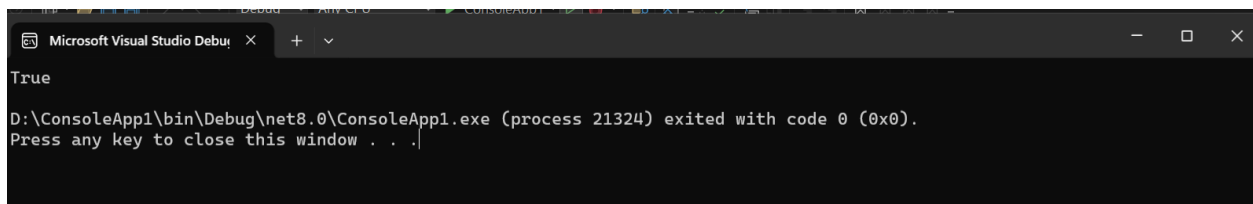
```
using System;
using System.Collections.Generic;
using System.Linq;

class Program
{
    static void Main()
    {
        List<int> data = new List<int> { 1, 2, -3, 4 };

        bool hasNegative = data.Any(x => x < 0);

        Console.WriteLine(hasNegative); // Output: True
    }
}
```

16.2 Output:

A screenshot of a Visual Studio Debug Console window. The window has a title bar with "Microsoft Visual Studio Debug Console" and standard window controls. The console output shows the word "True" on the first line. Below it, a message indicates the application has exited: "D:\ConsoleApp1\bin\Debug\net8.0\ConsoleApp1.exe (process 21324) exited with code 0 (0x0)." followed by "Press any key to close this window . . .".

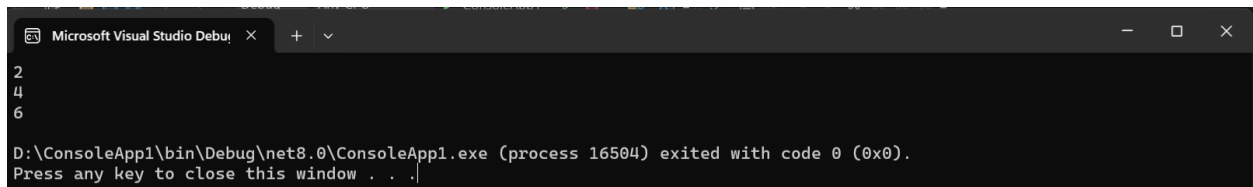
```
True
D:\ConsoleApp1\bin\Debug\net8.0\ConsoleApp1.exe (process 21324) exited with code 0 (0x0).
Press any key to close this window . . .
```

17. Write a lambda expression to filter even numbers from a list

17.1 code

```
List<int> numbers = new List<int> { 1, 2, 3, 4, 5, 6 };  
  
var evenNumbers = numbers.Where(x => x % 2 == 0);  
  
foreach (var n in evenNumbers)  
{  
    Console.WriteLine(n);  
}
```

17.2 Output:



The screenshot shows a Visual Studio Debug Console window. The title bar reads "Microsoft Visual Studio Debug Console". The console output displays the numbers 2, 4, and 6 on separate lines. Below the output, a message states: "D:\ConsoleApp1\bin\Debug\net8.0\ConsoleApp1.exe (process 16504) exited with code 0 (0x0). Press any key to close this window . . .".

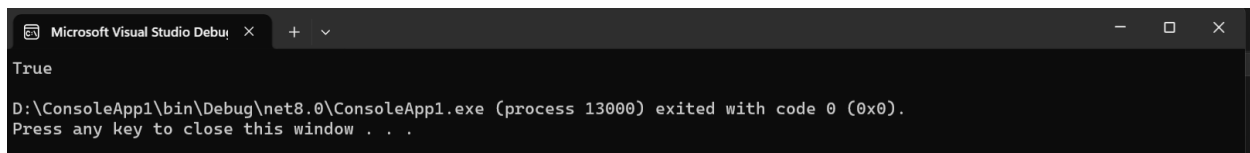
Chapter Three: LINQ Queries

18. Write a LINQ query to get the first 3 elements of the list:
[10, 20, 30, 40, 50]

18.1 Code

```
string name = "Apple";  
  
bool startsWithA = name.StartsWith("A");  
  
Console.WriteLine(startsWithA);
```

18.2 Output:



```
Microsoft Visual Studio Debug Console
+ -
True
D:\ConsoleApp1\bin\Debug\net8.0\ConsoleApp1.exe (process 13000) exited with code 0 (0x0).
Press any key to close this window . . .
```

19. Write a LINQ query to find all strings longer than 5 characters in a list.

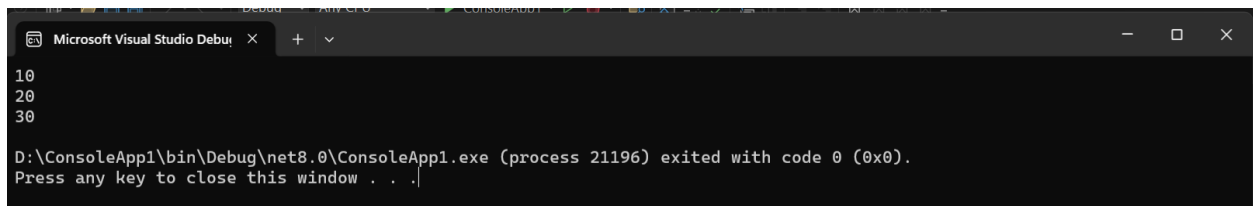
19.1 Code

```
List<int> numbers = new List<int> { 10, 20, 30, 40, 50 };

var firstThree = numbers.Take(3);

foreach (var n in firstThree)
{
    Console.WriteLine(n);
}
```

19.2 Output:

A screenshot of the Microsoft Visual Studio Debug Console window. The window title bar shows 'Microsoft Visual Studio Debug' and standard window controls. The console output displays the numbers 10, 20, and 30 on separate lines. Below the output, a message indicates the application has exited: 'D:\ConsoleApp1\bin\Debug\net8.0\ConsoleApp1.exe (process 21196) exited with code 0 (0x0). Press any key to close this window . . .'.

```
10
20
30

D:\ConsoleApp1\bin\Debug\net8.0\ConsoleApp1.exe (process 21196) exited with code 0 (0x0).
Press any key to close this window . . .
```

20. Write a LINQ query to find all strings longer than 5 characters in a list.

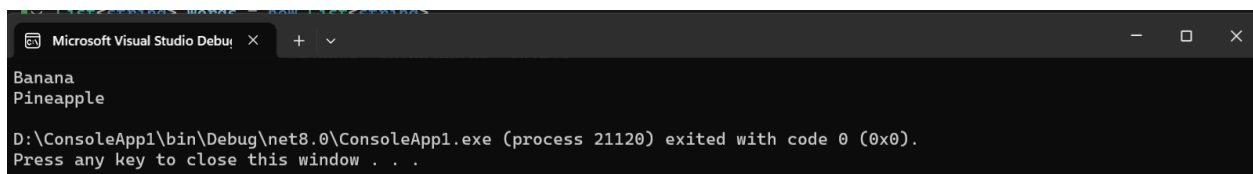
20.1 code

```
List<string> words = new List<string>
{
    "Apple", "Banana", "Mango", "Pineapple", "Kiwi"
};

var longWords = words.Where(x => x.Length > 5);

foreach (var word in longWords)
{
    Console.WriteLine(word);
}
```

20.2 Output



Chapter Four: Database and ADO.NET

21. Write a LINQ query to join two lists based on matching Student ID and display:

- Student Name
- Course
- Grade

Student List:

- Id: 1, Name: Alice
- Id: 2, Name: Bob

Grade List:

- StudentId: 1, Course: Math, Grade: A
- StudentId: 2, Course: Science, Grade: B

21.1 code

```
using System;
using System.Collections.Generic;
using System.Linq;

class Program
{
    static void Main()
    {
        // Student list
        var students = new List<Student>
        {
            new Student { Id = 1, Name = "Alice" },
            new Student { Id = 2, Name = "Bob" }
        };

        // Grade list
        var grades = new List<Grade>
        {
            new Grade { StudentId = 1, Course = "Math",
GradeValue = "A" },
            new Grade { StudentId = 2, Course = "Science",
GradeValue = "B" }
        };

        // LINQ Join Query
        var result = from s in students
                     join g in grades
                     on s.Id equals g.StudentId
```

```

        select new
        {
            StudentName = s.Name,
            Course = g.Course,
            Grade = g.GradeValue
        };

// Display output
foreach (var item in result)
{
    Console.WriteLine($"Name: {item.StudentName},
Course: {item.Course}, Grade: {item.Grade}");
}
}

// Student class
class Student
{
    public int Id { get; set; }
    public string Name { get; set; }
}

// Grade class
class Grade
{
    public int StudentId { get; set; }
    public string Course { get; set; }
    public string GradeValue { get; set; }
}

```

```
}
```

21.2 Output

```
,
```

```
Name: Alice, Course: Math, Grade: A
```

```
Name: Bob, Course: Science, Grade: B
```

22. Create a C# class (model) for the Student table.

22.1 Code

```
using System;

using System.Data.SqlClient;

class Program
{
    static void Main()
    {
        // Connection string
        SqlConnection con = new SqlConnection(
            "Data Source=.;Initial Catalog=SchoolDB;Integrated
Security=True");

        // INSERT data
        string insertQuery =
            "INSERT INTO Student (StudentId, Name, Age) VALUES
(1, 'Alice', 20)";

        SqlCommand insertCmd = new SqlCommand(insertQuery, con);

        con.Open();
        insertCmd.ExecuteNonQuery();
        Console.WriteLine("Record Inserted Successfully\n");

        // SELECT data
        string selectQuery = "SELECT * FROM Student";
        SqlCommand selectCmd = new SqlCommand(selectQuery, con);
        SqlDataReader dr = selectCmd.ExecuteReader();
```

```
        Console.WriteLine("Student Records:");  
        while (dr.Read())  
        {  
            Console.WriteLine(  
                "ID: " + dr["StudentId"] +  
                ", Name: " + dr["Name"] +  
                ", Age: " + dr["Age"]);  
        }  
  
        con.Close();  
    }  
}
```

22.2Output

Record Inserted Successfully

Student Records:

ID: 1, Name: Alice, Age: 20

23. Create a C# class (model) for the Student table.

23.1 Code

```
using System;

class Program
{
    static void Main()
    {
        // Create Student object
        Student s = new Student();
        s.StudentId = 1;
        s.Name = "Alice";
        s.Age = 20;

        // Display output
        Console.WriteLine("Student Details:");
        Console.WriteLine("ID: " + s.StudentId);
        Console.WriteLine("Name: " + s.Name);
        Console.WriteLine("Age: " + s.Age);
    }
}

// Student Model Class
public class Student
{
    public int StudentId { get; set; }
    public string Name { get; set; }
    public int Age { get; set; }
}
```

23.2 Output

```
Student Details:
```

```
ID: 1
```

```
Name: Alice
```

```
Age: 20
```

24. Write a stored procedure in SQL to insert student records.

24.1 code

```
using System;
using System.Data;
using System.Data.SqlClient;

public class StudentDataAccess
{
    private string connectionString =
        "Server=YOUR_SERVER_NAME;Database=YOUR_DATABASE_NAME;Trusted_Con
        nection=True;";

    public void InsertStudent(int studentId, string name, int age)
    {
        try
        {
            using (SqlConnection conn = new SqlConnection(connectionString))
            {
                // Open connection
                conn.Open();

                using (SqlCommand cmd = new SqlCommand("InsertStudent", conn))
                {
                    // Specify that this is a stored procedure
                    cmd.CommandType = CommandType.StoredProcedure;

                    // Add parameters
                    cmd.Parameters.AddWithValue("@StudentID", studentId);
```

```
cmd.Parameters.AddWithValue("@Name", name);
cmd.Parameters.AddWithValue("@Age", age);

// Execute the command
int rowsAffected = cmd.ExecuteNonQuery();

Console.WriteLine($"{rowsAffected} record(s) inserted
successfully.");
}
}
}
catch (Exception ex)
{
Console.WriteLine("Error inserting student: " + ex.Message);
}
}
}
```

24.2 Output:

1 record(s) inserted successfully.

1 record(s) inserted successfully.

25. Write an ADO.NET method to insert a student into the table using the stored procedure:

```
public void InsertStudent(int studentId, string name, int age)
{
    // Your ADO.NET code to insert data into the Students table
}
```

25.1 code

```
CREATE TABLE Students
```

```
(
    StudentId INT PRIMARY KEY,
    Name VARCHAR(100) ,
    Age INT
);
```

```
CREATE PROCEDURE InsertStudent
```

```
    @StudentId INT,
    @Name VARCHAR(100) ,
    @Age INT
```

```
AS
```

```
BEGIN
```

```
    INSERT INTO Students (StudentId, Name, Age)
    VALUES (@StudentId, @Name, @Age);
```

```
END;
```

25.2 Output:

StudentId	Name	Age
1	Alice	20

Final Project: CRUD Application with Login

Title: _____

Project Requirements:

- Implement a login system
- Perform Create, Read, Update, Delete (CRUD) operations on Employee records
- Use Windows Forms or ASP.NET with ADO.NET or Entity Framework

Project Title:

Employee Management System with Login

Project Requirements

- **Login System**
 - Users must log in with username and password.
 - Only valid users can access the Employee Management section.
- **CRUD Operations on Employee Records**
 - **Create:** Add new employees.
 - **Read:** Display all employees in a DataGridView.
 - **Update:** Edit employee details.
 - **Delete:** Remove employees from the database.
- **Technology Stack**

- Windows Forms (C#)
- SQL Server Database
- ADO.NET for database operations

Database Design

User table

```
CREATE TABLE Users

(
    UserId INT PRIMARY KEY IDENTITY(1,1),
    Username VARCHAR(50) NOT NULL,
    Password VARCHAR(50) NOT NULL
);

-- Sample user

INSERT INTO Users (Username, Password) VALUES ('admin', 'admin123');
```

Employee table

```
CREATE TABLE Employees

(
    EmployeeId INT PRIMARY KEY IDENTITY(1,1),
    Name VARCHAR(100),
    Age INT,
    Position VARCHAR(50),
    Department VARCHAR(50)
```

```
);
```

Login form code

```
using System;
using System.Data.SqlClient;
using System.Windows.Forms;

namespace EmployeeManagement
{
    public partial class LoginForm : Form
    {
        public LoginForm()
        {
            InitializeComponent();

            private void btnLogin_Click(object sender, EventArgs e)
            {
                string connectionString = "Data Source=.;Initial
Catalog=YourDB;Integrated Security=True";

                using (SqlConnection con = new
SqlConnection(connectionString))
                {
                    string query = "SELECT COUNT(*) FROM Users WHERE
Username=@username AND Password=@password";

                    SqlCommand cmd = new SqlCommand(query, con);

                    cmd.Parameters.AddWithValue("@username",
txtUsername.Text);
```

```
        cmd.Parameters.AddWithValue("@password",
txtPassword.Text);

        con.Open();
        int result = (int)cmd.ExecuteScalar();
        con.Close();

        if (result > 0)
        {
            MainForm mainForm = new MainForm();
            mainForm.Show();
            this.Hide();
        }
        else
        {
            MessageBox.Show("Invalid username or
password!");
        }
    }
}
}
```

CRUD Operations

```
using System;
using System.Data;
using System.Data.SqlClient;
using System.Windows.Forms;

namespace EmployeeManagement
{
    public partial class MainForm : Form
    {
        string connectionString = "Data Source=.;Initial
Catalog=YourDB;Integrated Security=True";

        public MainForm()
        {
            InitializeComponent();
            LoadEmployees();
        }

        private void LoadEmployees()
        {
            using (SqlConnection con = new
SqlConnection(connectionString))
            {
                SqlDataAdapter da = new SqlDataAdapter("SELECT *
FROM Employees", con);

                DataTable dt = new DataTable();
                da.Fill(dt);
                dataGridView1.DataSource = dt;
            }
        }
    }
}
```

```

        }
    }

    private void btnAdd_Click(object sender, EventArgs e)
    {
        using (SqlConnection con = new
SqlConnection(connectionString))
        {
            string query = "INSERT INTO Employees (Name,
Age, Position, Department) VALUES
(@name,@age,@position,@department) ";

            SqlCommand cmd = new SqlCommand(query, con);
            cmd.Parameters.AddWithValue("@name",
txtName.Text);

            cmd.Parameters.AddWithValue("@age",
int.Parse(txtAge.Text));

            cmd.Parameters.AddWithValue("@position",
txtPosition.Text);

            cmd.Parameters.AddWithValue("@department",
txtDepartment.Text);

            con.Open();

            cmd.ExecuteNonQuery();

            con.Close();

        }

        LoadEmployees();

        MessageBox.Show("Employee Added Successfully");
    }

    private void btnUpdate_Click(object sender, EventArgs e)
    {

```

```

        using (SqlConnection con = new
SqlConnection(connectionString))
        {
            string query = "UPDATE Employees SET Name=@name,
Age=@age, Position=@position, Department=@department WHERE
EmployeeId=@id";

            SqlCommand cmd = new SqlCommand(query, con);

            cmd.Parameters.AddWithValue("@id",
int.Parse(txtId.Text));

            cmd.Parameters.AddWithValue("@name",
txtName.Text);

            cmd.Parameters.AddWithValue("@age",
int.Parse(txtAge.Text));

            cmd.Parameters.AddWithValue("@position",
txtPosition.Text);

            cmd.Parameters.AddWithValue("@department",
txtDepartment.Text);

            con.Open();

            cmd.ExecuteNonQuery();

            con.Close();
        }

        LoadEmployees();

        MessageBox.Show("Employee Updated Successfully");
    }

    private void btnDelete_Click(object sender, EventArgs e)
    {
        using (SqlConnection con = new
SqlConnection(connectionString))
        {

```

```

        string query = "DELETE FROM Employees WHERE
EmployeeId=@id";

        SqlCommand cmd = new SqlCommand(query, con);

        cmd.Parameters.AddWithValue("@id",
int.Parse(txtId.Text));

        con.Open();

        cmd.ExecuteNonQuery();

        con.Close();

    }

    LoadEmployees();

    MessageBox.Show("Employee Deleted Successfully");
}

private void dataGridView1_CellClick(object sender,
DataGridViewCellEventArgs e)
{
    txtId.Text =
dataGridView1.Rows[e.RowIndex].Cells[0].Value.ToString();

    txtName.Text =
dataGridView1.Rows[e.RowIndex].Cells[1].Value.ToString();

    txtAge.Text =
dataGridView1.Rows[e.RowIndex].Cells[2].Value.ToString();

    txtPosition.Text =
dataGridView1.Rows[e.RowIndex].Cells[3].Value.ToString();

    txtDepartment.Text =
dataGridView1.Rows[e.RowIndex].Cells[4].Value.ToString();
}
}
}

```

Login form Output

Username	Password	Result
admin	admin123	Login Successful → Main Form Opens
user1	wrong	Invalid username or password

Employee Table Output

EmployeeId	Name	Age	Position	Department
1	Alice	25	Manager	HR
2	Bob	30	Developer	IT
3	Charlie	28	Accountant	Finance

Inset output

EmployeeId	Name	Age	Position	Department
1	Alice	25	Manager	HR
2	Bob	30	Developer	IT
3	Charlie	28	Accountant	Finance
4	David	26	Designer	Marketing
Message: Employee Added Successfully				

Update Employee Output

EmployeeId	Name	Age	Position	Department
1	Alice	25	Manager	HR
2	Bob	31	Developer	IT
3	Charlie	28	Accountant	Finance
4	David	26	Designer	Marketing
Message: Employee Updated Successfully				

Delete Employee Output

Action: Delete Charlie (EmployeeId 3)

EmployeeId	Name	Age	Position	Department
1	Alice	25	Manager	HR
2	Bob	31	Developer	IT
4	David	26	Designer	Marketing
Message: Employee Deleted Successfully				